

10 (long)

(a) The  $x'y'z'$  axes are obtained by rotating the  $xyz$  axes by an angle  $45^\circ$  anticlockwise about the  $y$  axis.

(i) Determine the matrix  $R$  such that the vector  $\mathbf{v}$  in the  $xyz$  system takes the form  $R\mathbf{v}$  in the  $x'y'z'$  system. [6]

(ii) Consider a matrix  $B$  in the  $xyz$  system. Explain why  $B$  becomes  $B' = RBR^T$  in the  $x'y'z'$  system. [5]

(b) The  $3 \times 3$  matrix  $A$  has eigenvalues  $\lambda_1, \lambda_2, \lambda_3$  and corresponding eigenvectors  $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$ . Determine the eigenvalues and corresponding eigenvectors of  $A^n$ , where  $n$  is a positive integer. [5]

(c) Consider the matrix

$$A = \begin{pmatrix} 10 & 0 & 0 \\ 0 & 3 & -2 \\ 0 & -2 & 3 \end{pmatrix}$$

(i) Calculate  $\lambda_1, \lambda_2, \lambda_3$  and  $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$ , the eigenvalues and eigenvectors of  $A$ . [8]

(ii) Calculate  $A^{10}\mathbf{x}$  for  $\mathbf{x} = (1, 2, 1)^T$ . [6]